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LLC Resonant Converter With Reconfigurable Secondary-Side for Output Voltage Regulation

Samuel S. Queiroz¹, Graduate Student Member, IEEE, and Levy F. Costa¹, Member, IEEE

Abstract—This brief proposes a reconfigurable LLC resonant converter that enables the regulation of a tailored portion of the output voltage through a novel adaptive transformer turns ratio scheme implemented on the secondary rectifier. The converter employs a center-tapped transformer and a switch to reconfigure the secondary rectifier and adjust the effective turns ratio. Using the proposed concept, voltage regulation is achieved with a reduced number of switches and devices in the current's path, no modification to the primary side, and fixed-frequency operation. A laboratory prototype rated at 4 kW with a 1 kV input voltage is implemented and tested to validate the feasibility and performance of the proposed converter.

Index Terms—DC, transformer, LLC resonant converter, output voltage regulation, reconfigurable rectifier.

I. INTRODUCTION

THE LLC resonant DC/DC converter is widely used in several applications due to its high efficiency, soft-switching capability, and simple implementation [1], [2]. It has become a popular choice for solid-state transformer (SST) and electric vehicles (EVs) charging applications [3], [4], especially when operating as a DC transformer, i.e., constant voltage gain and output voltage [5], [6]. In this operating mode, the converter achieves high efficiency, as demonstrated in [7], but it inherently lacks output voltage regulation.

In SSTs, only limited output regulation (around 5-20%) is typically required to compensate for potential input voltage sags, whereas EV charging stations require a wide voltage range (e.g., 400–800 V). To regulate the output voltage of the LLC resonant converter as a DC transformer, different techniques have been discussed in the literature, including modulation strategies and topological reconfiguration.

Modulation-based approaches, such as phase-shift modulation and pulse-width modulation (PWM) [8], [9], [10], will increase switching losses, which significantly reduce efficiency, mainly in high-power applications, which is not desired in SST systems. Another approach is the topological reconfiguration techniques, which are based on the modification of the circuit configuration on the primary and/or secondary side to achieve output voltage adjustment. For instance, in [11] and [12], reconfigurable voltage quadrupler rectifiers are proposed for applications requiring a wide output voltage

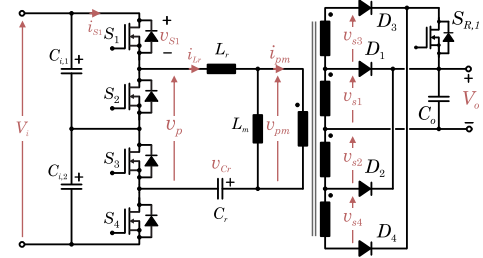


Fig. 1. Architecture of the proposed LLC resonant converter.

range. It is capable of doubling or quadrupling the output voltage, depending on the rectifier configuration. However, the main drawback of these solutions is the large step in the output voltage variation (twice or four times the base voltage), besides the need for additional hardware, such as an active switch and multiple capacitors. Similarly, a reconfigurable converter capable of modifying the primary bridge operation and the secondary side rectifier is proposed in [13]. This converter also has the drawback of using additional hardware (e.g., bidirectional switch), fixed step voltage variation, and increased current stress on the primary side devices, which compromises efficiency. Finally, in [14] and [15], reconfigurable rectifiers are proposed based on modifications to the transformer structure, requiring additional components such as bidirectional switches, diodes and capacitors. The extra hardware and higher number of devices in the path of the current (compromising the efficiency) are remarkable drawbacks.

In this context, this brief proposes a novel LLC resonant converter using a reconfigurable rectifier with an adjustable transformer turns ratio, as shown in Fig. 1. Different from previous solutions presented in the literature, the proposed converter enables regulation of a tailored portion of the output voltage by adjusting the transformer turns ratio through the reconfigurable secondary-side rectifier. Unlike doubler or quadrupler converters, the voltage step in the proposed topology is not constrained to fixed values. Instead, it can be tailored according to the application, enabling small steps (e.g., 5-20%) suitable for SST systems, or larger steps such as output voltage doubling for EV charging stations, which typically require voltage regulation between 400-800 V. In other words, the output voltage step is a design choice. Even though the proposed converter introduces an additional switch on the secondary side (denoted as $S_{R,1}$ in Fig. 1), the overall hardware remains simpler compared to other solutions, leading to a reduced number of active components in the current path. Moreover, the switch $S_{R,1}$ is unidirectional, which simplifies

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The authors are with the Department of Electrical Engineering, Eindhoven University of Technology, 5612 AP Eindhoven, The Netherlands (e-mail: s.soares.queiroz@tue.nl).

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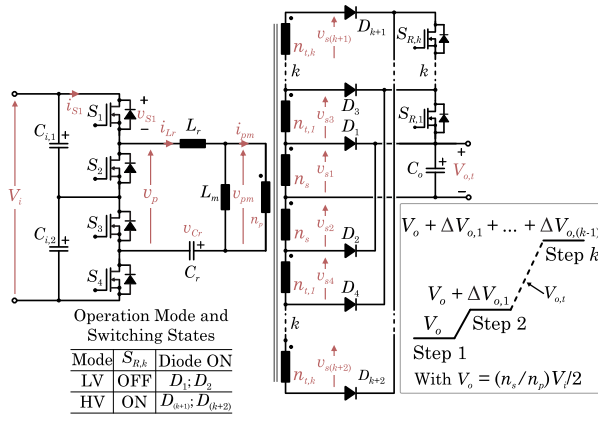


Fig. 2. Generalized architecture of the proposed LLC resonant converter.

the gate driver circuitry and reduces conduction losses, unlike previous solutions that require bidirectional switches. Furthermore, when the switch $S_{R,1}$ is turned on, only one additional semiconductor is added in the current path, with no extra diodes. In this brief, the operating principle of the proposed topology is presented in Section II, followed by the theoretical analysis and design details in Section III. Experimental results are then provided in Section IV to demonstrate the effectiveness of the proposed converter.

II. PROPOSED RECONFIGURABLE LLC RESONANT CONVERTER AND OPERATING PRINCIPLE

The proposed reconfigurable LLC resonant converter is depicted in Fig. 1 and can be generalized to offer more output voltage steps, as illustrated in Fig. 2. The converter comprises a three-level (3L) half-bridge (HB) on the primary side [16], [17], a resonant tank (resonant inductor L_r , resonant capacitor C_r and magnetizing inductor L_m), a center-tapped medium-frequency transformer (MFT) with reconfigurable rectifier on the secondary side. The proposed secondary bridge consists of the center-tapped windings n_s , $2k$ auxiliary windings $n_{t,k}$, $2k$ diodes and $(k-1)$ unidirectional active switches, where k represents the number of output voltage steps. In the generalized topology, each auxiliary winding adds a voltage step $\Delta V_{o,k}$ when its corresponding active switch $S_{R,k}$ is turned on. The added voltage is defined by $\Delta V_{o,k} = (V_i/2)(n_{t,k}/n_p)$. Therefore, the generalized output voltage of the proposed converter expressed as

$$V_{o,t} = \frac{n_s}{n_p} \frac{V_i}{2} + \sum_{i=1}^k \frac{V_i}{2} \frac{n_{t,i}}{n_p}. \quad (1)$$

The auxiliary windings affect the overall transformer size and must be carefully considered during the MFT design. Their effect on the converter's operation is discussed in the next section. The output voltage behavior is illustrated in the diagram of Fig. 2, where each switch $S_{R,k}$ increases the output voltage by $\Delta V_{o,k}$, synthesizing the tailored steps and $V_{o,t}$. The LLC resonant converter operates as a DC transformer, with the switching frequency f_s set equal to or slightly below the resonant frequency f_o . The duty cycle of the primary switches driving signals is fixed at 0.5 [5]. The operation principle of the converter, shown in Fig. 1, is described in Fig. 3. The

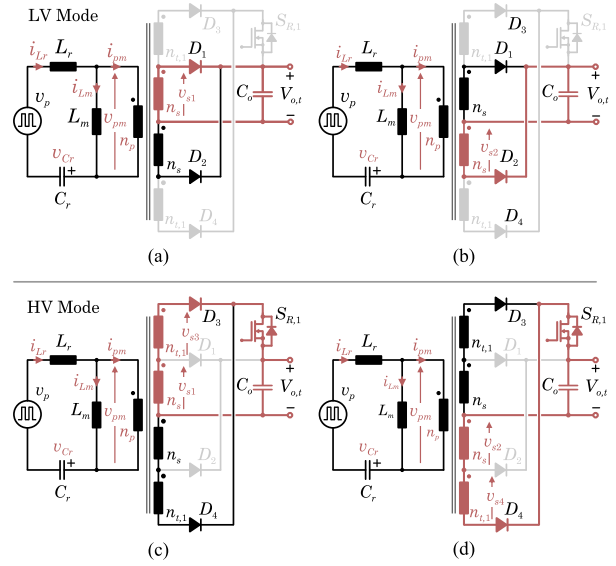


Fig. 3. Operation modes of the proposed reconfigurable LLC resonant converter: (a) - (b) LV mode; (c) - (d) HV mode.

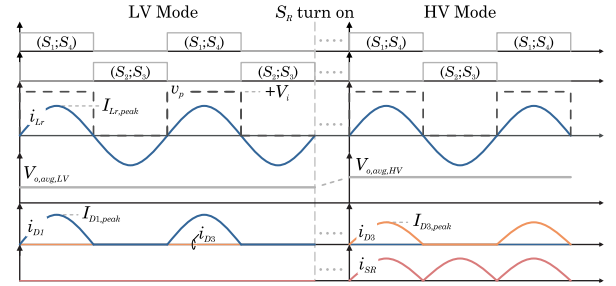


Fig. 4. Key waveforms of the reconfigurable LLC resonant converter in LV and HV modes.

converter operates in two distinct modes: low voltage (LV) mode and high voltage (HV) mode, as depicted in Fig. 3.

1) *LV Mode*: In this mode, the switch $S_{R,1}$ is permanently off, consequently the diodes D_3 and D_4 are reverse-biased and remain blocked, as illustrated in Fig. 3(a) and (b). During the positive half-cycle of the resonant current i_{Lr} , the primary voltage v_p is $+V_i$. The secondary current i_s flows through the upper half of the secondary winding and the diode D_1 , while the diode D_2 is reverse-biased, as shown in Fig. 3(a). The output voltage V_o is reflected to the primary side and L_m is clamped to V_o/n_{Tr} , where n_{Tr} is the transformer turns ratio. During the negative half-cycle of i_{Lr} , v_p becomes zero. The current i_s flows through lower half of secondary winding and the diode D_2 , as can be observed in Fig. 3(b). In LV mode, the n_{Tr} is given by n_s/n_p , resulting a voltage $V_{o(LV)} = V_i \cdot n_s/n_p$. The main waveforms for the LV mode are shown in Fig. 4.

2) *HV Mode*: In this mode, the switch $S_{R,1}$ is permanently on, and the diodes D_1 and D_2 are reverse-biased (i.e., out of the circuit). Therefore, the secondary rectifier operates using diodes D_3 and D_4 , as shown in Fig. 3(c) and (d). With S_R conducting, the entire secondary winding of the MFT becomes active, increasing the number of secondary turns. During the positive half-cycle of i_{Lr} , diode D_3 is forward-biased, and i_s flows through the full upper secondary winding, as depicted in Fig. 3(c). In this case, the V_o is given by $v_{s1} + v_{s3}$. Similarly, for the negative half-cycle, diode D_4 conducts and i_s flows

through the lower secondary winding, resulting in an V_o of $v_{s2} + v_{s4}$, as shown in Fig. 3(d). In HV mode, n_{Tr} is $(n_s + n_{t,1})/n_p$, and $V_{o(HV)} = V_i \cdot (n_s + n_{t,1})/n_p$. The main waveforms for the HV mode are shown in Fig. 4.

III. ANALYSIS AND DESIGN OF THE RECONFIGURABLE LLC RESONANT CONVERTER

The LLC resonant converter is designed to operate in discontinuous conduction mode (DCM) within the inductive region. The switching frequency is set at or slightly below the resonant frequency (i.e., $f_s \leq f_o$). When operating near f_o , the converter achieves its highest efficiency, with zero-voltage switching (ZVS) for the primary side switches and zero current switching (ZCS) for the secondary-side diodes, while also minimizing the circulating current. The resonant tank is composed of L_r , C_r and L_m , which define the resonant frequency as $f_o = 1/(2\pi \cdot \sqrt{L_r C_r})$. As shown in Fig. 4, the 3L HB converter generates v_p with two levels: 0 and $+V_i$. Thus, the maximum MFT primary voltage is $V_i/2$. However, a dc offset of $V_i/2$ appears across C_r . The sinusoidal current i_{Lr} is defined as $I_{Lr,peak} \cdot \sin(\omega_o t)$, where the peak current $I_{Lr,peak}$ is determined by $\frac{n_{Tr} P_o}{4 f_s V_o \sqrt{L_r C_r}}$, with P_o representing the nominal power. The secondary current in the diode is derived from i_{Lr} and its peak value is given by $I_{Lr,peak}/n_{Tr}$ (see Fig. 4).

A. Voltage Gain

To determine the voltage gain M_v , the first harmonic approximation can be used, as described in detail in [5]. The normalized M_v of the LLC resonant converter is given by

$$M_v = \frac{f_n^2 L_n}{\sqrt{[f_n^2(L_n + 1) - 1]^2 + [f_n L_n Q_e(f_n^2 - 1)]^2}}, \quad (2)$$

where M_v is determined in terms of the inductor ratio $L_n = L_m/L_r$, quality factor $Q_e = \frac{\pi^2 n_{Tr}^2}{8 R_o} \cdot \sqrt{\frac{L_r}{C_r}}$ and relative frequency $f_n = f_s/f_o$. In the proposed topology, the effective n_{Tr} changes according to the operation mode. In the LV mode, only part of the secondary winding is active. The effective turns ratio is $n_{Tr(LV)} = \frac{n_s}{n_p} = \frac{V_{o,LV}}{V_i/2}$. For the HV mode, the full secondary winding is active and the effective turns ratio is $n_{Tr(HV)} = \frac{n_s + n_{t,1}}{n_p} = \frac{V_{o(HV)}}{V_i/2}$. Therefore, the corresponding voltage gain is described as $G_{v(LV)} = n_{Tr(LV)} \cdot M_v$ for LV mode and $G_{v(HV)} = n_{Tr(HV)} \cdot M_v$ for HV mode.

B. Design Methodology and Effect of the Topological Reconfiguration

In the LLC resonant converter design, L_r is often realized using the MFT leakage inductance, typically referred to the primary side [6]. However, in the proposed converter, the leakage inductance varies according to the operating mode. To clarify this phenomenon, a simplified MFT model is derived and shown in Fig. 5(a), where the primary and secondary leakages are represented by $L_{l,p}$ and $L_{l,s}$, respectively. In LV mode, only the secondary windings $n_{s,1}$ and $n_{s,2}$ are connected, resulting in a total secondary leakage inductance of $L_{S,eq(LV)} = L_{l,s1} + L_{l,s2}$. In contrast, in HV mode, all four

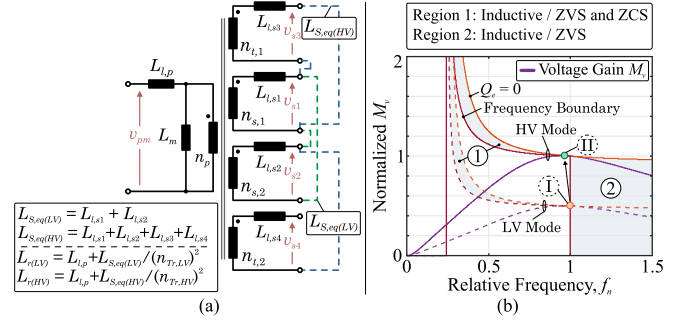


Fig. 5. (a) Equivalent model of the MFT in the reconfigurable LLC resonant converter; (b) normalized voltage gain of the converter.

secondary windings are active, and the total secondary leakage inductance is given by $L_{S,eq(HV)} = L_{l,s1} + L_{l,s2} + L_{l,t1} + L_{l,t2}$. Therefore, the equivalent resonant inductance referred to the primary is $L_{r(LV)} = L_{l,p} + L_{S,eq(LV)}/(n_{Tr(LV)}^2)$, for LV mode, and $L_{r(HV)} = L_{l,p} + L_{S,eq(HV)}/(n_{Tr(HV)}^2)$, for HV mode. It is important to note that the turns ratio in the HV mode is higher than LV mode ($n_{Tr(HV)} > n_{Tr(LV)}$), resulting in a lower equivalent resonant inductance in HV mode compared to LV mode ($L_{r(HV)} < L_{r(LV)}$). In contrast, the capacitance C_r remains the same in both modes, since the capacitor bank does not change. Note also that L_m is independent of the MFT configuration. The rated power of the converter is the same regardless the operation mode, resulting in the output load in LV and HV modes being $R_{o(LV)} < R_{o(HV)}$. Based on these parameters, the quality factor for each mode can be derived as

$$Q_e = \begin{cases} \frac{\pi^2 n_{Tr(LV)}^2}{8 R_{o(LV)}} \cdot \sqrt{\frac{L_{r(LV)}}{C_r}}, & \text{LV Mode} \\ \frac{\pi^2 n_{Tr(HV)}^2}{8 R_{o(HV)}} \cdot \sqrt{\frac{L_{r(HV)}}{C_r}}, & \text{HV Mode.} \end{cases} \quad (3)$$

In the design of the LLC resonant converter, a higher Q_e narrows the resonance peak of the M_v curve. Considering the converter parameters, we find $Q_{e(LV)} > Q_{e(HV)}$. As a result, the resonant tank must be designed based on the LV mode, as it presents the highest quality factor and lowest inductor ratio, which significantly influences the converter's operation and soft-switching performance, representing the worst-case scenario. Accordingly, the capacitor C_r is calculated using $L_{r(LV)}$ by means of $1/(4\pi^2 f_o^2 L_{r(LV)})$. As the converter is designed based on the LV mode configuration ($L_r = L_{r(LV)}$), it operates at point I within region 1, as depicted in Fig. 5(b). When it switches from LV to HV mode, the resonant inductance decreases to $L_{r(HV)}$, increasing f_o . Consequently, the f_n drops below unity and the converter shifts to operation point II, which also lies in region 1, as shown in Fig. 5(b). Thus, the converter maintains full soft switching and inductive. If instead the design were based on HV mode parameters, switching to LV mode would decrease f_o , resulting in $f_n > 1$, and the operation in region 2 (see Fig. 5(b)), where soft switching on the secondary side would be lost. This behavior is an intrinsic characteristic of the proposed reconfigurable topology and is critical for both the converter design and operation.

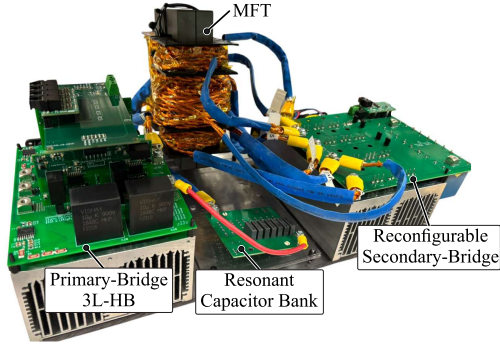


Fig. 6. Picture of the implemented prototype.

TABLE I
SPECIFICATIONS AND COMPONENTS OF THE RECONFIGURABLE
LLC RESONANT CONVERTER

Specifications	Value	Components	
Input Voltage	$V_i = 1 \text{ kV}$	MOSFET $S_{1-4} / S_{R,1}$	C3M0016120K
Output Voltage	$V_o = 350\text{--}600 \text{ V}$	Diode	NDSH40120CDN
Nominal Power	$P_N = 4 \text{ kW}$	Turns Ratio	$n_{Tr,LV} = 0.7$ $n_{Tr,HV} = 1.222$
Switching Frequency	$f_s = 50 \text{ kHz}$	MFT	$L_m = 2 \text{ mH}$ $L_{r(LV)} = 146 \mu\text{H}$ $L_{r(HV)} = 132 \mu\text{H}$

IV. EXPERIMENTAL RESULTS

A prototype was built to verify the effectiveness and performance of the proposed converter, with the specifications listed in Table I. Fig. 6 shows a picture of the experimental prototype. As explained in the last section, the converter is designed based on the LV mode. As the converter is designed to operate as a DC transformer, the f_s is defined slightly below f_o , thus $f_s = 49 \text{ kHz}$ and $f_o = 50 \text{ kHz}$. From the capacitor equation, it is found $C_r = 0.7 \text{ nF}$.

Fig. 7(a)–(d) presents the main steady-state waveforms of the proposed reconfigurable LLC resonant converter in the LV mode. The current i_{Lr} and voltages V_o , V_p and v_{cr} are presented in Fig. 7(a). The waveforms behave as expected and the converter regulates V_o at 350 V. The behavior of the voltage V_{S1} over S_1 and the current i_{S1} through it are presented in Fig. 7(b), where it is demonstrated the ZVS operation in the primary bridge. The waveforms of the currents i_{D1} and i_{D2} are shown in Fig. 7(c), whereas the voltage over D_1 and D_2 are presented in Fig. 7(d). As can be seen, the diodes in the LV mode handle the voltage $V_{o(LV)} + v_{s1}$ and the ZCS is achieved. Similarly, Fig. 8(f)–(h) shows the main steady-state waveforms of the converter in the HV mode, with i_{Lr} , i_{D3} and i_{D4} in Fig. 8(g) and i_{D3} alongside the voltages over D_1 , D_3 and D_4 in Fig. 8(h). It is important to note that the voltage V_o is regulated at 600 V and V_{D1} remains negative throughout HV mode, indicating that diode D_1 is continuously reverse-biased. Furthermore, the diodes in HV mode withstand the voltage $2V_{o(HV)}$. The experimental results of the reconfigurable LLC resonant converter are consistent with the theoretical waveforms, validating the operation of the proposed converter in both modes. Additionally, it demonstrates that the developed

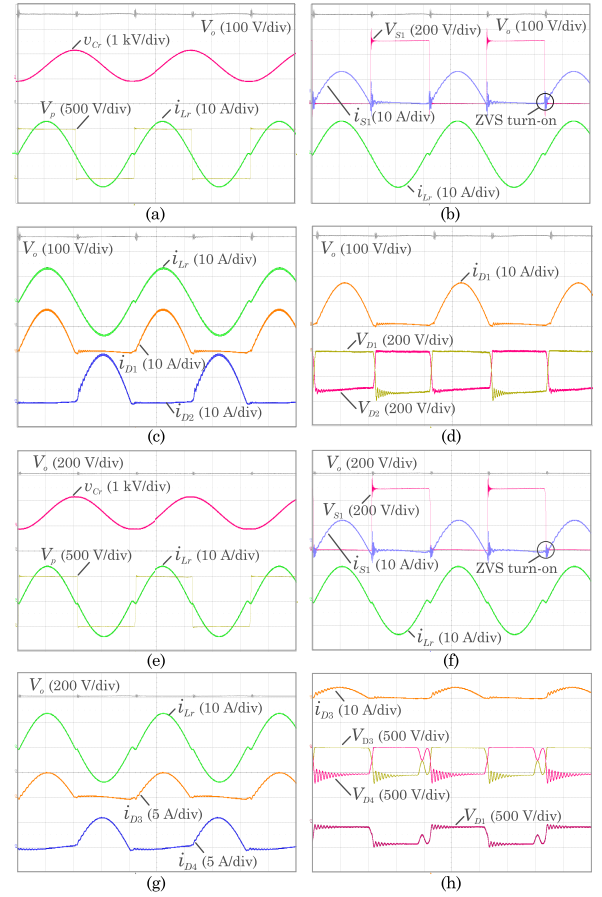


Fig. 7. Experimental waveforms: (a)–(d) LV mode - $V_{o(LV)} = 350 \text{ V}$; (e)–(h) HV mode - $V_{o(HV)} = 620 \text{ V}$.

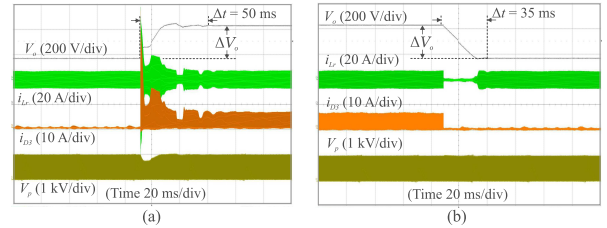


Fig. 8. Dynamic waveforms of the proposed converter: (a) transition from LV to HV mode; (b) transition from HV to LV mode.

design methodology ensures soft-switching in the primary and secondary bridges in both operation modes.

The transition mode waveforms at $P_o = 2 \text{ kW}$ are presented in Fig. 8. Fig. 8(a) shows the behavior of i_{Lr} , V_o , i_{D3} and V_p in the dynamic transition from LV to HV mode. It is clear that V_o changes from 350 V to 600 V with a transient duration of approximately 50 ms. Fig. 8(b) illustrates the waveforms in the transition from HV to LV mode, where V_o changes from 600 V to 350 V with a transient duration of about 35 ms. The current i_{Lr} is constant, indicating the constant power operation.

The efficiency of the proposed LLC resonant converter was evaluated in both modes. As shown in Fig. 9(a), the converter achieves 96.7% in LV mode and 97.6% in HV mode at $P_o = 4 \text{ kW}$. Overall, the converter demonstrates higher efficiency in HV mode. Fig. 9(b) shows the loss breakdown at $P_o = 4 \text{ kW}$. Primary-side losses remain constant, as the primary current does not depend on V_o . The total losses are dominated

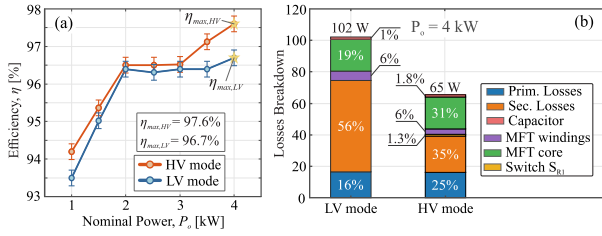


Fig. 9. (a) Measured efficiency in function of the output power; (b) power losses breakdown in LV and HV modes.

TABLE II
COMPARISON WITH STATE-OF-THE-ART

Parameter	[10]	[11]	[12]	[13]	[14]	Proposed
Sw. Freq. [kHz]	100	35	50–100	150–250	140	50
Switches	5	5	5	6	4	5
Diodes	6	4	4	4	6	4
Bi-dir. Switches	0	0	0	0	2	0
Capacitors	5	6	5	2	2	2
Active Elements*	6	7	7	5	6	3
V_i [V]	390	150–400	400	400	390	1000
V_o [V]	250–420	1650	100–800	150–450	60–450	350–600
P_o [kW]	1.3	2.5	0.5	3.3	1.8	4
V_o Steps**	Fixed	Fixed	Fixed	Fixed	Fixed	Scalable
η_{peak} [%]	94	99.1	95.5	96.9	97.5	97.6

* The number of active elements refers to the maximum number of switches, diodes, and capacitors in the current path in the secondary bridge during the converter's operation.

** The V_o steps refer to the amplitude of the output voltage levels generated by the reconfiguration of the topology.

by secondary conduction losses. The efficiency difference is mainly due to higher secondary current in LV mode, leading to increased conduction losses in diodes and windings. Moreover, in HV mode, the losses of the switch S_{R1} are very low (around 1.3%) and have a minor impact on the efficiency.

Table II compares the proposed converter with existing solutions. It achieves fixed frequency soft-switching operation over a wide output voltage range, while requiring fewer devices overall and the lowest number of active elements in the current path. Unlike prior designs with fixed output voltage steps, the proposed topology is inherently scalable, allowing tailored step magnitudes in the output voltage defined by the secondary-side auxiliary windings. With only one additional secondary switch, it offers a good trade-off between simplicity, scalability, and functionality, reaching a competitive efficiency of 97.6%.

V. CONCLUSION

This brief proposed a novel reconfigurable LLC resonant converter using a center-tapped transformer and a secondary-side reconfigurable rectifier. The converter delivers fixed output voltage steps through a secondary-side active switch while maintaining constant switching frequency. Experimental results from a 1 kV / 4 kW prototype confirm the feasibility, effective voltage regulation, and turns ratio adaptation of the design. The converter also shows good dynamic performance in the transition between the LV and HV modes and high efficiency in both modes. Furthermore, the converter attains a peak efficiency of 97.6% in HV mode at 4 kW. Despite the extra losses from the active switch in the HV mode, the overall

semiconductor losses are reduced due to the lower current stress on the secondary side.

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